Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    int count = 0;

    // Read input (no counter increment per instructions)
    scanf("%d", &n);

    // Initial assignment: rev = 0
    int rev = 0;
    int remainder;
    count++; // count for rev = 0 assignment

    // Loop condition checks: n != 0
    while (count++, n != 0) {

        // remainder = n % 10
        remainder = n % 10;
        count++;

        // rev = rev * 10 + remainder
        rev = rev * 10 + remainder;
        count++;

        // n /= 10
        n /= 10;
        count++;
    }

    // print(rev) statement execution
    count++;

    // Print counter variable (no counter increment per instructions)
    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔